

Surname

Forename(s)

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

MATHEMATICS**4008/1**

PAPER 1

JUNE 2013 SESSION

2 hours 30 minutes

Candidates answer on the question paper.

Additional materials:

Geometrical instruments

Allow candidates 5 minutes to count pages before the examination.

TIME 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page and your Centre number and Candidate number on the top right corner of every page of this paper.

If you need additional working space, use the lined pages at the back and number your work correctly.

Answer all questions.

Check that all the pages are in the booklet and ask the invigilator for a replacement if there are duplicate or missing pages.

Write your answers in the spaces provided on the question paper using black or blue pens.

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Mathematical tables, slide rules and calculators should not be brought into the examination room.

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The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

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ZIMSEC 12013

Turn over

NEITHER MATHEMATICAL TABLES NOR SLIDE RULES NOR CALCULATORS MAY BE USED IN THIS PAPER.

(a) Subtract -2 from 2 .

(b) Leaving your answer as a common fraction, find, in its lowest terms, the value of $\frac{8}{15} + \frac{2}{3}$.

Suggested solution

(a) $2 - (-2) = 2 + 2 = 4$

(b) $\frac{8}{15} + \frac{2}{3} = \frac{8}{15} + \frac{4}{3} = \frac{8}{15} + \frac{20}{15} = \frac{28}{15}$

Answer	(a)	4	[1]
	(b)	$\frac{28}{15}$	[2]

(a) Express $3\frac{4}{5}$ as a decimal number.

(b) Find the exact value of $\frac{0,8 + 8,368}{0,42}$.

Suggested solution

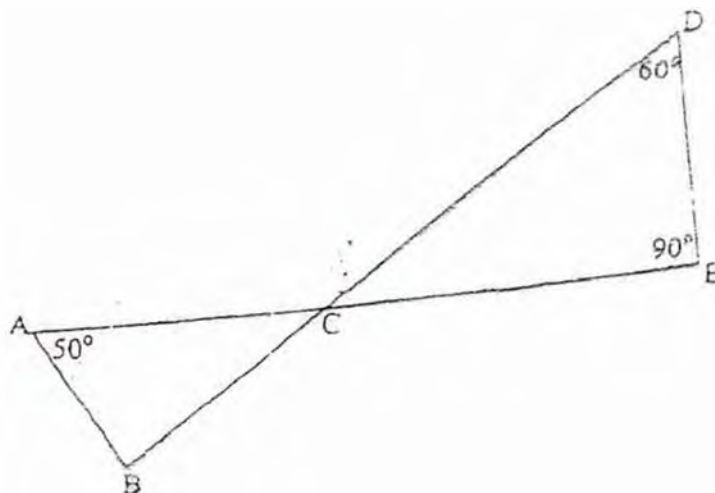
(a) $3\frac{4}{5} = \frac{19}{5} = 3,8$

(b)
$$\begin{array}{r} 0,830 \\ + 8,368 \\ \hline 9,198 \end{array}$$

$\frac{9,198}{0,42} = \frac{919,8}{42} = 21\frac{9}{10}$

Answer	(a)	3,8	[1]
	(b)	21,9	[2]

- 3 (a) Find n such that $0.0075 = 7.5 \times 10^n$
 (b)



In the diagram ACE and BCD are straight lines intersecting at C. Given that $\angle CED = 90^\circ$, calculate $\angle ABC$.

Suggested solution

3 (a) $0.0075 = 7.5 \times 10^n$

$7.5 \times 10^{-3} = 7.5 \times 10^n$,

comparing powers of 10.

$n = -3$

(b) $\angle ACB = 180^\circ - (90^\circ + 60^\circ) = 30^\circ$

$\angle ABC = 180^\circ - (50^\circ + 30^\circ) = 100^\circ$

Answer (a) $n = -3$ [1]
 (b) 100° [2]

- 4 (a) Write down the square of 4.

- (b) Evaluate $125^{\frac{1}{3}} \times \sqrt{144}$.

Suggested solution

4 (a)

$4^2 = 4 \times 4 = 16$

(b)

$125^{\frac{1}{3}} \times \sqrt{144} = (5^3)^{\frac{1}{3}} \times 12$

$= 5^{3 \times \frac{1}{3}} \times 12$

$= 5^1 \times 12$

$= 60$

Answer (a) 16 [1]
 (b) 60 [2]

Express

(a) 3 m^2 in cm^2 .

(b) $32,5 \text{ m/s}$ in km/h .

but $1 \text{ m} = 100 \text{ cm}$

expected solution

(a)

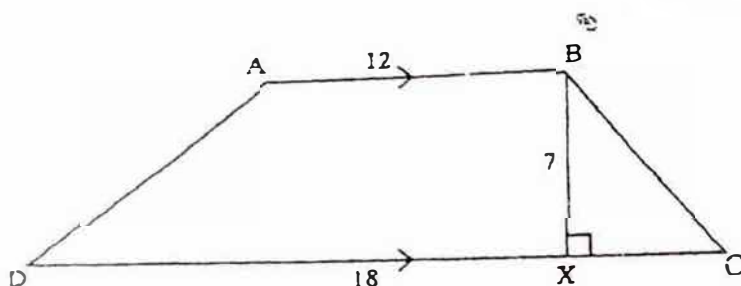
$$\begin{aligned} 3 \text{ m}^2 &= 3 \times \text{m} \times \text{m}, \\ &= 3 \times 100 \text{ cm} \times 100 \text{ cm} \\ &= 30\,000 \text{ cm}^2 \end{aligned}$$

(b)

$$\begin{aligned} 32,5 \text{ m/s} &= \frac{32,5 \text{ m}}{\text{s}} \\ &= \frac{32,5}{1000} \times \frac{3600}{1} \text{ km/h} \\ &= 11,7 \text{ km/h} \end{aligned}$$

Answer (a) $30\,000 \text{ cm}^2$ [1]
(b) $11,7 \text{ km/h}$ [2]

6



In the diagram, ABCD is a quadrilateral in which AB is parallel to DC, $AB = 12 \text{ cm}$, $CD = 18 \text{ cm}$, $BX = 7 \text{ cm}$ and $\angle BXC = 90^\circ$.

(a) State the special name given to the quadrilateral ABCD.

(b) Calculate the area of the quadrilateral.

expected solution

(a)

TRAPEZIUM

(b)

$$\begin{aligned} \text{Area} &= \frac{1}{2} (AB + CD) \times BX \\ &= \frac{1}{2} \times (12 \text{ cm} + 18 \text{ cm}) \times 7 \text{ cm} \\ &= \frac{1}{2} \times (30 \text{ cm}) \times 7 \text{ cm} \\ &= 105 \text{ cm}^2 \end{aligned}$$

Answer (a) Trapezium

[1]

7 Simplify $\frac{x^2 + 7x + 6}{x^2 - 36}$.

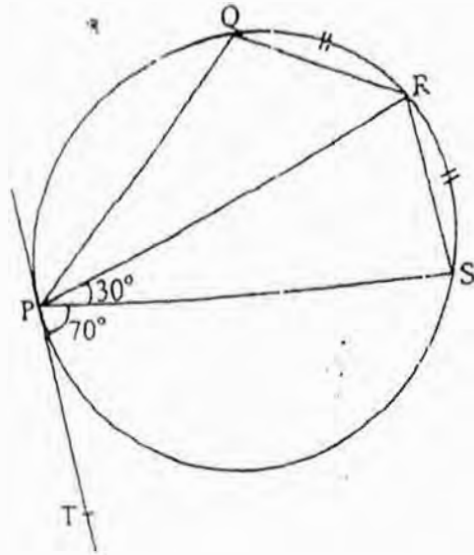
Suggested solution

$$\begin{aligned} 7 \quad \frac{x^2 + 7x + 6}{x^2 - 36} &= \frac{x^2 + x + 6x + 6}{x^2 - 6^2} = \frac{x(x+1) + 6(x+1)}{(x-6)(x+6)} \\ &= \frac{(x+6)(x+1)}{(x-6)(x+6)} \\ &= \frac{(x+1)}{(x-6)} \end{aligned}$$

Answer

$$\frac{(x+1)}{(x-6)}$$

[3]



In the diagram P, Q, R and S are points on the circumference of a circle and arcs QR and RS are equal. TP is a tangent to the circle at P. $\angle TPQ = 70^\circ$ and $\angle RPS = 30^\circ$

- Calculate
- $\angle QPR$,
 - $\angle PRS$,
 - $\angle PQR$.

Worked solution

- | | | |
|-----|---|---|
| (a) | $\angle QPR = 30^\circ$ | (Δ 's subtended by equal chords & arcs) |
| (b) | $\angle PRS = \angle TPS = 70^\circ$ | (Δ in alternate segment) |
| (c) | $\angle PQR = \angle TPR$
$= 70^\circ + 30^\circ$
$= 100^\circ$ | (Δ in alternate segment) |

Answer	(a)	30°	[1]
	(b)	70°	[1]
	(c)	100°	[1]

9 E varies directly as the square of V .

(a) Express E in terms of V and a constant m .

(b) Given that $E = 3$ when $V = 2$ find m .

Suggested solution

9 (a) $E \propto V^2 \Rightarrow E = mV^2$, where m is a constant

(b) When $V = 2$, $E = 3$ we get

$$3 = m \times 2^2$$

$$3 = 4m$$

$$m = \frac{3}{4}$$

Answer	(a)	$E = mV^2$	[1]
	(b)	$m = \frac{3}{4}$	[2]

10 Evaluate (a) $(-3)^0$,

(b) $\left(\frac{16}{81}\right)^{\frac{3}{4}}$.

Suggested solution

10 (a) $(-3)^0 = 1$

(b) $\left(\frac{16}{81}\right)^{\frac{3}{4}} = \left(\frac{81}{16}\right)^{\frac{3}{4}}$

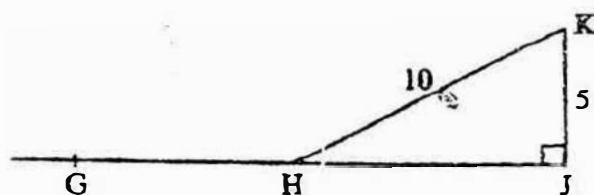
$$= \frac{81^{\frac{3}{4}}}{16^{\frac{3}{4}}}$$

$$= \frac{(\sqrt[4]{81})^3}{(\sqrt[4]{16})^3}$$

$$\begin{aligned}
 &= \frac{(3)^3}{(2)^3} \\
 &= \frac{3 \times 3 \times 3}{2 \times 2 \times 2} \\
 &= \frac{27}{8} \\
 &= 3\frac{3}{8}
 \end{aligned}$$

Answer	(a)	1	{1}
	(b)	$3\frac{3}{8}$	{2}

11



In the diagram GHJ is a straight line. $\hat{HJK} = 90^\circ$, JK = 5 cm and HK = 10 cm.

- (a) Find $\sin \hat{GHK}$,
 (b) Calculate HJ leaving your answer in surd form.

Suggested solution

11 (a)

$$\sin \hat{GHK} = \sin (180^\circ - \hat{KHJ}) = \sin \hat{KHJ}$$

$$= \frac{5}{10}$$

$$= \frac{1}{2}$$

(b)

$$HJ^2 + JK^2 = HK^2,$$

using Pythagoras theorem

$$HJ^2 + 5^2 = 10^2$$

$$HJ^2 + 25 = 100$$

$$HJ^2 = 75$$

$$HJ = \sqrt{75}$$

$$= \sqrt{25 \times 3}$$

$$= \sqrt{25} \times \sqrt{3}$$

(a)

$$3x + 2y = 18$$

$$2y = -3x + 18$$

$$y = -\frac{3}{2}x + \frac{18}{2}$$

$$y = -\frac{3}{2}x + 9$$

Gradient, $m = -\frac{3}{2}$

(b)

Method 1

Recall that parallel lines have equal gradients

Using, $y = mx + c$, with $x = -2, y = 3$ and $m = -\frac{3}{2}$

i.e. $3 = -\frac{3}{2}(-2) + c \Rightarrow 3 = 3 + c \Rightarrow c = 0$

$$y = -\frac{3}{2}x$$

$$\Rightarrow 3x + 2y = 0$$

Method 2

Using, $\frac{y - y_1}{x - x_1} = m$, with $(x_1, y_1) = (-2, 3)$ and $m = -\frac{3}{2}$

i.e. $\frac{y - 3}{x - (-2)} = -\frac{3}{2}$

$$\frac{y - 3}{x + 2} = -\frac{3}{2}$$

$$2(y - 3) = -3(x + 2)$$

$$2y - 6 = -3x - 6$$

$$3x + 2y = 6 - 6$$

$$3x + 2y = 0$$

Answer (a) $-\frac{3}{2}$ or $-1\frac{1}{2}$ [1]

(b) $3x + 2y = 0$ [2]

14 Given that $h\begin{pmatrix} 3 \\ 5 \end{pmatrix} + k\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 14 \\ 6 \end{pmatrix}$, find the scalars h and k .

Suggested solution

14

$$h\begin{pmatrix} 3 \\ 5 \end{pmatrix} + k\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 14 \\ 6 \end{pmatrix}$$

$$\begin{pmatrix} 3h \\ 5h \end{pmatrix} + \begin{pmatrix} 2k \\ -k \end{pmatrix} = \begin{pmatrix} 14 \\ 6 \end{pmatrix}$$

$$\begin{pmatrix} 3h + 2k \\ 5h - k \end{pmatrix} = \begin{pmatrix} 14 \\ 6 \end{pmatrix}$$

$$= 5 \times \sqrt{3}$$

$$= 5\sqrt{3}$$

Answer (a) $\frac{1}{2}$ [1]
(b) $5\sqrt{3}$ [2]

12 Given that $M = \begin{pmatrix} 5 & 5 \\ 3 & x \end{pmatrix}$ and $N = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

- find (a) the determinant of M in terms of x ,
(b) the modulus of the vector N ,
(c) the value of x given that $\det M = 15$.

Suggested solution

12 (a) $\det(M) = 5x - 5 \times 3 = 5x - 15$

(b) $|N| = \sqrt{3^2 + 4^2}$
 $= \sqrt{9 + 16}$
 $= \sqrt{25}$
 $= 5$

(c) $\det M = |N|$
 $5x - 15 = 5$
 $x - 3 = 1$
 $x = 4$

divide both sides by 5 to solve the equation.

Answer (a) $\det M = 5x - 15$ [1]
(b) $|N| = 5$ [1]
(c) $x = 4$ [4]

- 13 (a) Write down the gradient of the line whose equation is $3x + 2y = 18$.
(b) Find the equation of the straight line which is parallel to the line $3x + 2y = 18$ and passes through $(-2; 3)$.

Suggested solution

Method 1 (Substitution Method)

$$\begin{array}{rcl} 3h + 2k & = & 14 \quad \text{---} \quad \textcircled{1} \\ 5h - k & = & 6 \quad \text{---} \quad \textcircled{2} \end{array}$$

From $\textcircled{2}$ $5h - k = 6$, or $5h - 6 = k$

Substituting into $\textcircled{1}$ gives,

$$3h + 2(5h - 6) = 14$$

$$3h + 10h - 12 = 14$$

$$13h = 14 + 12$$

$$13h = 26$$

$$h = 2$$

from $\textcircled{1}$

$$3(2) + 2k = 14$$

$$6 + 2k = 14$$

$$3 + k = 7$$

$$k = 4$$

Method 2 (Elimination Method)

$$\begin{array}{rcl} 3h + 2k & = & 14 \quad \text{---} \quad \textcircled{1} \times 1 \\ 5h - k & = & 6 \quad \text{---} \quad \textcircled{2} \times 2 \end{array}$$

$$3h + 2k = 14 \quad \text{---} \quad \textcircled{3}$$

$$10h - 2k = 12 \quad \text{---} \quad \textcircled{4}$$

$\textcircled{3} + \textcircled{4}$:

$$13h = 26$$

$$h = 2$$

Using $\textcircled{2}$

$$3(2) + 2k = 14$$

$$6 + 2k = 14$$

$$3 + k = 7$$

$$k = 4$$

Answer	$h = 2$	$[2]$
	$k = 4$	$[2]$

15 Given that $\log_{10} 3 = 0,4771$ and $\log_{10} 5 = 0,6991$, find

(a) $\log_{10} 1\frac{2}{3}$,

(b) $\log_{10} 30$.

Suggested solution

15 (a) $\log_{10} 1\frac{2}{3} = \log_{10} \frac{5}{3}$,

$$= \log_{10} 5 - \log_{10} 3$$

$$= 0,6991 - 0,4771$$

$$= 0,222$$

but $\log\left(\frac{A}{B}\right) = \log A - \log B$

(b) $\log_{10} 30 = \log_{10}(3 \times 10)$

recalling that $\log AB = \log A + \log B$

$$= \log_{10} 3 + \log_{10} 10$$

$$= 0,4771 + 1$$

Since $\log_{10} 10 = 1$

$$= 1,4771$$

Answer

(a) 0,222

(b) 1,4771

21 A trader bought a tonne of goods worth \$2 500.

- (a) Calculate the cost price per kilogram.
- (b) If the goods were later sold at \$2.10 per kilogram. Calculate the percentage loss.
- (c) Find the value of $n^4 - 4n$ if $n = 3$

Suggested solution

21 (a) $\text{Cost/kg} = \frac{2\,500}{1\,000}$
 $= \$2.50$

(b) Amount at \$2.10/kg $= \$2.10 \times 100$
 $= \$2100$

$$\begin{aligned}\text{Percentage loss} &= \frac{\$2\,500 - \$2\,100}{\$2\,500} \times 100\% \\ &= \frac{4}{25} \times 100\% \\ &= 16\%\end{aligned}$$

(c) $n^4 - 4n = 3^4 - 4(3)$ given that $n = 3$
 $= 3 \times 3 \times 3 \times 3 - 12$
 $= 81 - 12$
 $= 69$

Answer	(a)	\$2.50	[1]
	(b)	16%	[2]
	(c)	69	[2]

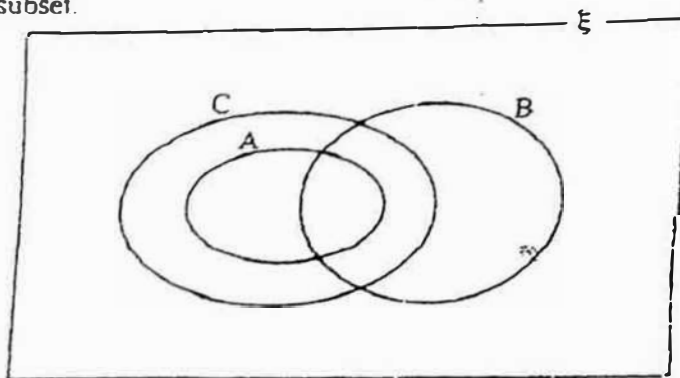
22 Given that $\xi = \{x : 1 \leq x \leq 15, x \text{ is an integer}\}$,

$$A = \{x : x \text{ is a multiple of 4}\},$$

$$B = \{x : x \text{ is a perfect square}\}$$

and $C = \{x : x \text{ is a multiple of 2}\}.$

- (a) In the Venn diagram in the working space, fill in the members of each subset.



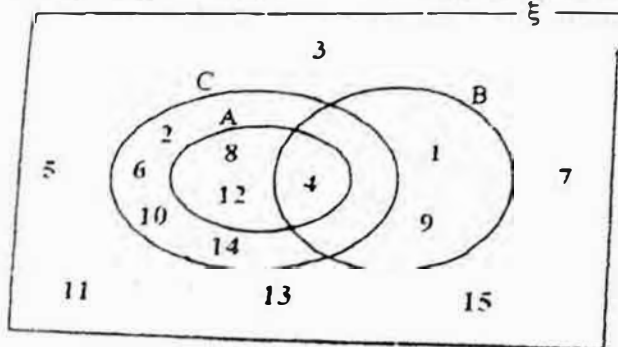
- (b) (i) Write down the relationship between sets A and C in set notation.
(ii) Find $n(A \cap B \cap C)$.

suggested solution

(a)

$$\xi = \{1; 2; 3; 4; 5; \dots; 15\}, \quad A = \{4; 8; 12\}$$

$$B = \{1; 4; 9\} \quad \text{and} \quad C = \{2; 4; 6; 8; 10; 12; 14\}$$



- (b) (i) $A \subseteq C$
(ii) $n(A \cap B \cap C) = 1$

Answer	(a) Venn diagram	[3]
	(b) (i) $A \subseteq C$	[1]
	(ii) $n(A \cap B \cap C) = 1$	[1]

- 23 A teacher gave ball-point pens as prizes to pupils who passed his test. He had 2 boxes of pens. Box A had 6 blue, 4 green and 3 red pens while Box B had 6 blue and 4 green pens. Ben was asked to pick a pen from Box A and Laiza from Box B.

Find the probability that

- (a) Ben picked a blue pen.
- (b) both Ben and Laiza picked blue pens.
- (c) both Ben and Laiza picked pens of the same colour.

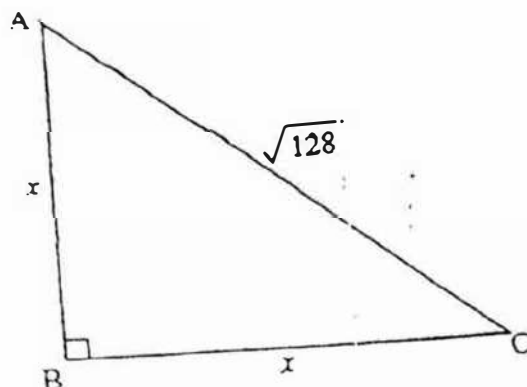
Suggested solution

- 23 (a) $P(\text{Ben pick blue}) = \frac{6}{13}$
- (b) $P(\text{Ben picked blue \& Laiza picked blue}) = P(\text{Ben picked blue}) \times P(\text{Laiza picked blue})$
- $$= \frac{6}{13} \times \frac{6}{10}$$
- $$= \frac{36}{130}$$
- $$= \frac{18}{65}$$
- (c) $P(\text{Both Ben \& Laiza picked pens of the same colours})$
- $$= P(\text{Ben picked blue \& Laiza picked blue}) + P(\text{Ben picked green \& Laiza picked green})$$
- $$= \frac{6}{13} \times \frac{6}{10} + \frac{4}{13} \times \frac{4}{10}$$
- $$= \frac{26}{65}$$
- $$= \frac{2}{5}$$

Answer	(a)	$\frac{6}{13}$	[1]
	(b)	$\frac{18}{65}$	[2]
	(c)		

- 24 (a) If n is an integer, calculate the greatest possible value of n which satisfies the inequality $3n - 25 < 2$.

(b)



In the diagram $AB = BC = x$ cm, $AC = \sqrt{128}$ cm and $\angle ABC = 90^\circ$.

- (i) Form an equation in x .
(ii) Find the value of x .

Suggested solution

24

(a)

$$3n - 25 < 2$$

$$3n < 2 + 25$$

$$3n < 27$$

$$n < 9$$

$$n = 8$$

(b) (i) $x^2 + x^2 = (\sqrt{128})^2$,

by Pythagoras theorem

$$2x^2 = 128$$

$$x^2 = 64$$

$$x^2 - 64 = 0$$

(ii) $x^2 - 64 = 0$

$$x^2 - 8^2 = 0$$

difference of two squares

$$(x - 8)(x + 8) = 0$$

Either $x - 8 = 0$ or $x + 8 = 0$

i.e. $x = -8$ or $x = 8$

but $x > 0$

Hence $x = 8$

Answer (a)

(b) (i)
(ii)

$$\begin{aligned} n &= 8 \\ x^2 - 64 &= 0 \\ x &= 8 \end{aligned}$$

[2]
[1]
[3]

25

A class of 40 pupils from different families were asked how many pets they kept. The results are shown in the table.

number of pets per family	1	2	3	4
number of families	3	7	17	13

- (a) (i) State the mode.
(ii) Calculate the mean number of pets per family.
(b) If the results in the table were shown on a pie chart, calculate the angle representing the number of pupils who kept two pets.
(c) Express the number of pupils who kept three pets as a percentage of the class.

Suggested solution

- 25 (a) (i) 3
(ii) Mean = $\frac{1 \times 3 + 2 \times 7 + 3 \times 17 + 4 \times 13}{40}$
 $= \frac{120}{40}$
 $= 3$
(b) Angle = $\frac{7}{40} \times 360 = 63^\circ$
(c) Percentage = $\frac{17}{40} \times 100\% = 42.5\%$

Answer	(a)	(i)	3	[1]
		(ii)	3	[3]
	(b)		63°	[2]
	(c)		42.5%	[3]

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General Certificate of Education Ordinary Level

4008/1

MATHEMATICS

PAPER 1

NOVEMBER 2013 SESSION

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NEITHER MATHEMATICAL TABLES NOR SLIDE RULES NOR CALCULATORS
MAY BE USED IN THIS PAPER.

1 Evaluate, giving each answer as a fraction in its lowest terms.

(a) $\frac{1}{5} + \frac{1}{6}$

(b) $\frac{2}{5} \div 4$

(c) $\frac{3}{4} - \frac{1}{4} \times \frac{2}{3}$

Suggested solution

(a) $\frac{1}{5} + \frac{1}{6} = \frac{6+5}{30} = \frac{11}{30}$

(b) $\frac{2}{5} \div 4 = \frac{2}{5} \div \frac{4}{1} = \frac{2}{5} \times \frac{1}{4}$
 $= \frac{1}{5} \times \frac{1}{2}$
 $= \frac{1}{10}$

(c) $\frac{3}{4} - \frac{1}{4} \times \frac{2}{3} = \frac{3}{4} - \left(\frac{1}{4} \times \frac{2}{3}\right)$
 $= \frac{3}{4} - \frac{1}{6}$
 $= \frac{9-2}{12}$
 $= \frac{7}{12}$

Answer

(a) $\frac{11}{30}$
 (b) $\frac{1}{10}$
 (c) $\frac{7}{12}$

Evaluate $\frac{(0.3)^3 \times 0.02}{0.0008}$, giving your answer in standard form.

Suggested solution

$$\frac{(0.3)^3 \times 0.02}{0.0008} = \frac{(0.3 \times 0.3 \times 0.3) \times 0.02}{0.0008}$$

$$= \frac{0.027 \times 0.02}{0.0008}$$

$$= \frac{0.00054}{0.0008}$$

$$= \frac{54}{8}$$

$$= 6.75$$

$$= 6.75 \times 1$$

$$= 6.75 \times 10^0$$

Answer 6.75×10^0 [3]

(a) The temperature inside a freezer is -8°C . During a power cut the temperature rose by 12°C . Find the temperature after the rise.

(b) Write down the next two terms in the following sequence; $1; \frac{1}{2}; \frac{1}{4}; \frac{1}{8}; \dots$

Suggested solution

(a) $-8^\circ + 12^\circ = 12^\circ - 8^\circ = 4^\circ$

(b) $1; \frac{1}{2}; \frac{1}{4}; \frac{1}{8}; \dots = \frac{1}{2^0}; \frac{1}{2^1}; \frac{1}{2^2}; \frac{1}{2^3}; \frac{1}{2^4}; \frac{1}{2^5}$

The next two terms are: $\frac{1}{2^4} = \frac{1}{2 \times 2 \times 2 \times 2} = \frac{1}{16}$ and $\frac{1}{2^5} = \frac{1}{2 \times 2 \times 2 \times 2 \times 2} = \frac{1}{32}$

Answer (a) 4° [1]

(b) $\frac{1}{16}; \frac{1}{32}$ [2]

4

(a) Write 3.35 minutes in minutes and seconds.

(b) If 1 kilometre is $\frac{5}{8}$ of a mile, convert 75 miles to kilometres.Suggested solution

$$\begin{aligned} \text{(a)} \quad 3.35 \text{ minutes} &= 3 \text{ minutes } \frac{35}{100} \times 60 \text{ seconds} \\ &= 3 \text{ minutes and } 21 \text{ seconds} \end{aligned}$$

(b) Let x be the number of kilometres

$$1 \text{ km} = \frac{5}{8} \text{ miles}$$

$$x = 75 \text{ miles}$$

$$\frac{x}{1} = \frac{75}{5/8}$$

$$x = \frac{75 \times 8}{5}$$

$$x = 120$$

Answer	(a)	3 minutes 21 seconds	11
	(b)	120 km	12

5

A shopper spent $\frac{c}{d}$ on one item and half of that amount on each of three other items.

Find how much she spent altogether.

Suggested solution

$$\begin{aligned} \frac{c}{d} + 3 \left(\frac{1}{2} \frac{c}{d} \right) &= \frac{c}{d} + \frac{3c}{2d} \\ &= \frac{2c + 3c}{2d} \\ &= \frac{5c}{2d} \end{aligned}$$

Answer	$\frac{5c}{2d}$
--------	-----------------

8

Simplify

(a) $\frac{(3^3)^4}{27^3}$

(b) $(4x^2y^6)^{\frac{1}{2}}$

(c) $x^0 + x^{-2}$

Suggested solution

(a)

$$\frac{(3^3)^4}{27^3} = \frac{(3 \times 3 \times 3)^4}{27^3}$$

$$= \frac{27^4}{27^3}$$

$$= 27^4 \div 27^3$$

$$= 27^{4-3}$$

$$= 27$$

using $(a \times b \times c)^d = a^d \times b^d \times c^d$

(b)

$$(4x^2y^6)^{\frac{1}{2}} = (4 \times x^2 \times y^6)^{\frac{1}{2}}$$

$$= 4^{\frac{1}{2}} \times (x^2)^{\frac{1}{2}} \times (y^6)^{\frac{1}{2}}$$

$$= (2^2)^{\frac{1}{2}} \times (x^2)^{\frac{1}{2}} \times (y^6)^{\frac{1}{2}}$$

$$= 2^{2 \times \frac{1}{2}} \times x^{2 \times \frac{1}{2}} \times y^{6 \times \frac{1}{2}}$$

$$= 2 \times x \times y^3$$

$$= 2xy^3$$

Method 1

(c)

$$x^0 \div x^{-2} = x^{0-(-2)}$$

$$= x^2$$

since $x^a \div x^b = x^{a-b}$ Method 2

$$x^0 \div x^{-2} = 1 \div \frac{1}{x^2}$$

$$= 1 \times \frac{x^2}{1}$$

$$= x^2$$

recall that $x^0 = 1$ and $x^{-a} = \frac{1}{x^a}$ **Answer**

(a) 27 [1]

(b) $2xy^3$ [1]

(c) x^2 [1]

7 (a) Express $\frac{7}{8}$ as a decimal fraction.

(b) A car loses 55% of its value after four years.

If it cost \$8 500 when new, find its value after the four years.

Suggested solution

7 (a) $\frac{7}{8} = 0.875$

(b) If a car loses 55% after 4 years, then its value after 4 years is 45%. Thus,

$$\begin{aligned} 45\% \text{ of } \$8\,500 &= \frac{45}{100} \times \$8\,500 \\ &= \$\frac{9}{20} \times 8\,500 \\ &= \$3\,825 \end{aligned}$$

Answer	(a)	0.875	[1]
	(b)	\$3 825	[2]

8 (a) Simplify $5m - 2(x - 3m)$.

(b) Solve the equation $\frac{x+5}{7} = \frac{3}{2}$.

Suggested solution

8 (a)
$$\begin{aligned} 5m - 2(x - 3m) &= 5m - 2x + 6m \\ &= 5m + 6m - 2x \\ &= 11m - 2x \end{aligned}$$

(b)
$$\begin{aligned} \frac{x+5}{7} &= \frac{3}{2} \\ 2(x+5) &= 7 \times 3 \\ 2x + 10 &= 21 \\ 2x &= 21 - 10 \\ 2x &= 11 \\ x &= \frac{11}{2} \end{aligned}$$

Answer	(a)	$11m - 2x$	[1]
	(b)	$x = 5\frac{1}{2}$ or 5.5	[2]

(b)

23 and 29

Answer (a) $x = -10$ or $x = 4$ [3]
(b) 23 and 29 [1]

13 (a) Simplify $\frac{x^2 + 3x + 2}{x + 2}$

(b) Find the order of rotational symmetry of a right-angled isosceles triangle.

Suggested solution

1.1 (a)
$$\begin{aligned}\frac{x^2 + 3x + 2}{x + 2} &= \frac{x(x + 1) + 2(x + 1)}{x + 2} \\ &= \frac{(x + 1)(x + 2)}{x + 2} \\ &= x + 1\end{aligned}$$

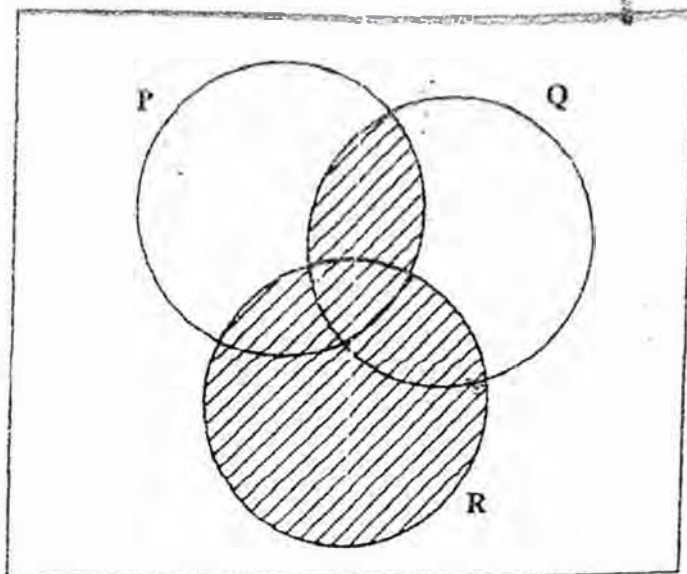
(b) Rotational order = 2

14 (a) Given that $n(A) = 10$ and $n(B) = 15$, find the greatest possible value of

(i) $n(A \cup B)$,

(ii) $n(A \cap B)$.

(b)



Use set notation to describe the shaded region in the above diagram in terms of sets P, Q and R.

Suggested solution

14

Answer (a) (i) 25

(ii) 10

(b) $(P \cap Q) \cup R$

15 Given that $A = \begin{pmatrix} -2 & -1 \\ 6 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & -1 \\ 4 & 3 \end{pmatrix}$.

Find (a) $3A - B$,

(b) B^2 .

Suggested solution

15 (a) $3A - B = 3 \begin{pmatrix} -2 & -1 \\ 6 & 2 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 4 & 3 \end{pmatrix}$
 $= \begin{pmatrix} 3 \times -2 & 3 \times -1 \\ 3 \times 6 & 3 \times 2 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 4 & 3 \end{pmatrix}$
 $= \begin{pmatrix} -6 & -3 \\ 18 & 6 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 4 & 3 \end{pmatrix}$
 $= \begin{pmatrix} -6 - 0 & -3 - (-1) \\ 18 - 4 & 6 - 3 \end{pmatrix} = \begin{pmatrix} -6 & -3 + 1 \\ 14 & 3 \end{pmatrix}$
 $= \begin{pmatrix} -6 & -2 \\ 14 & 3 \end{pmatrix}$

(b) $B^2 = B \times B$
 $= \begin{pmatrix} 0 & -1 \\ 4 & 3 \end{pmatrix} \times \begin{pmatrix} 0 & -1 \\ 4 & 3 \end{pmatrix}$
 $= \begin{pmatrix} 0 + (-1 \times 4) & (0 \times -1) + (-1 \times 3) \\ (4 \times 0) + (3 \times 4) & (4 \times -1) + (3 \times 3) \end{pmatrix}$
 $= \begin{pmatrix} 0 - 4 & 0 - 3 \\ 0 + 12 & -4 + 9 \end{pmatrix}$
 $= \begin{pmatrix} -4 & -3 \\ 12 & 5 \end{pmatrix}$

Answer (a) $\begin{pmatrix} -6 & -2 \\ 14 & 3 \end{pmatrix}$ [2]

(b) $\begin{pmatrix} -4 & -3 \\ 12 & 5 \end{pmatrix}$ [2]

Write down the largest four-digit number in base eight.

(b) Convert 111_8 to a number in base two.

(c) Find the sum of 444_5 and 21_5 giving your answer in base five.

Suggested solution

16 (a) 7777_8

$$(b) \quad 8^3 \times 1 + 8^2 \times 1 + 8^1 \times 1 = 64 \times 1 + 8 \times 1 + 1 \times 1$$

$$= 64 + 8 + 1$$

$$= 73_{10}$$

2	73		
2	36	r	1
2	18	r	0
2	9	r	0
2	4	r	1
2	2	r	0
2	1	r	0
	0	r	1

(c)

$$\begin{array}{r} 444_5 \\ + 21_5 \\ \hline 1020_5 \end{array}$$

Answer	(a)	7777_8	[1]
	(b)	1001001_2	[2]
	(c)	1020_5	[1]

- 17 A rugby team scored the following points in 12 matches 21; 18; 3; 12; 15; 18; 42; 18; 24; 6; 12; 3. For the 12 matches,

find

- (a) the mode.
- (b) the mean.
- (c) In the next match, the team scored 55 points.

Write down the median score for the 13 matches.

Suggested solution

- 17 (a) Rearranging the points in ascending order gives

3; 3; 6; 12; 15; 18; 18; 18; 21; 24; 42

The mode = 18

- (b) The mean = $\frac{3+3+6+12+12+12+15+18+18+18+21+24+42}{12}$

$$= \frac{192}{12}$$

$$= 16$$

- (c) 3; 3; 6; 12; 12; 15; 18; 18; 18; 21; 24; 42; 55

The median = 18

Answer	(a)	18	[1]
	(b)	16	[2]
	(c)	18	[1]

- 18 The scale of a map is 1:10 000.

- (a) Two hills are 4.5 cm apart on the map.

Find the actual distance between the hills, giving your answer in kilometres.

- (b) Two towns are 80 km apart.

Find the distance between them on the map, giving your answer in centimetres.

Suggested solution

- 18 (a) $1 \text{ km} = (1000 \times 100) \text{ cm}$

$$= 100\,000\text{ cm}$$

$$1\text{ cm} : 10\,000\text{ cm}$$

$$4,5\text{ cm} = \text{more}$$

$$= \frac{4,5}{1} \times 10\,000\text{ cm}$$

$$= 45\,000\text{ cm}$$

by simple proportion

$$\text{But } 1\text{ km} = 100\,000\text{ cm}$$

$$\text{less} = 45\,000\text{ cm}$$

$$= \frac{45\,000}{100\,000} \times 1\text{ km}$$

$$= 0,45\text{ km}$$

$$(b) \quad 0,1\text{ km} = 1\text{ cm}$$

$$80\text{ km} = x$$

$$0,1x = 80$$

$$x = 800\text{ cm}$$

Answer (a)	0,45 km	[2]
------------	---------	-----

(b)	800 cm	[2]
-----	--------	-----

19 The temperature $T^\circ\text{C}$ at a height of H metres above sea level, is given by the formula $T = 20 - \frac{H}{150}$.

(a) Calculate the temperature at 4 500 metres.

(b) Make H the subject of the formula.

(c) Find the height at which the temperature is 12°C .

Suggested solution

19 (a) $T = 20 - \frac{H}{150}$, where $H = 4\,500$

$$T = 20 - \frac{4\,500}{150}$$

$$T = 20 - 30$$

$$T = -10^\circ\text{C}$$

(b) $T = 20 - \frac{H}{150}$

$$T + \frac{H}{150} = 20$$

$$\frac{H}{150} = 20 - T$$

$$H = 150 \times (20 - T)$$

(c)

$$H = 150 \times (20 - 12)$$

$$= 150 \times 8$$

$$= 1\,200$$

answer

(a) -10°C [1]

(b) $H = 150(20 - T)$ [2]

(c) 1200 metres [1]

20

The number of revolutions, n , of a wheel over a fixed distance varies inversely as the circumference, C cm, of the wheel.

(a) Write down an equation involving n , C and a constant k .

(b) If a wheel of circumference 80 cm makes 10 revolutions, find the number of revolutions made by a wheel of circumference 200 cm.

Suggested solution

20 (a) $n \propto \frac{1}{C} \Rightarrow n = \frac{k}{C}$, where k is a constant.

(b) When $C = 80$, $n = 10$,

$$10 = \frac{k}{80}$$

$$10 \times 80 = k$$

$$k = 800$$

Thus, $n = \frac{800}{C}$

When $C = 200$, $n = n$

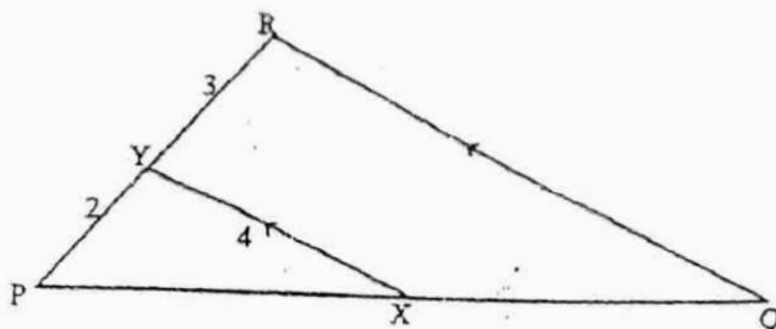
Hence, $n = \frac{800}{200} \Rightarrow n = 4$

Answer

(a) $n = \frac{k}{C}$ [1]

(b) $n = 4$ [3]

21



In the diagram, XY is parallel to QR , $PY = 2$ cm, $YR = 3$ cm and $XY = 4$ cm.

Find

(a) the length of QR ,

(b) the ratio $\frac{\text{area } \triangle PXY}{\text{area } \triangle PQR}$.

Suggested solution

21 (a) $\triangle PQR \sim \triangle PXY \quad \therefore \frac{QR}{XY} = \frac{PR}{PY}$ by definition of ratio of corresponding sides.

Hence, $\frac{QR}{4} = \frac{5}{2}$ or $QR = 10$

(b) $\frac{\text{area of } \triangle PXY}{\text{area of } \triangle PQR} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$

Answer:	(a)	10 cm	[2]
	(b)	$\frac{4}{25}$	[2]

- 22 Tendai and Vimbai take a driving test. The probability that Tendai will pass is $\frac{3}{5}$ and the probability that Vimbai will pass is $\frac{2}{3}$.

- (a) State which one of them is more likely to pass.
- (b) Calculate the probability that
- they both fail,
 - only one of them will pass.

Suggested solution

- 22 (a) $P(\text{Tendai will pass}) = \frac{3}{5} = 0.6$ and $P(\text{Vimbai will pass}) = \frac{2}{3} = 0.666666 \dots = 0.6$

Clearly, Vimbai is more likely to pass.

- (b) (i) $P(\text{Both fail}) = P(\text{Tendai fail}) \times P(\text{Vimbai fail})$

$$= \left(1 - \frac{3}{5}\right) \times \left(1 - \frac{2}{3}\right)$$

$$= \frac{2}{5} \times \frac{1}{3}$$

$$= \frac{2}{15}$$

- (ii) $P(\text{Only one will pass}) =$

$$= (P(\text{Tendai passes and Vimbai fail or Tendai fail and Vimbai passes})$$

$$= (P(\text{Tendai passes and Vimbai fail}) + (P(\text{Tendai fail and Vimbai passes}))$$

$$= \left(\frac{3}{5}\right)\left(\frac{1}{3}\right) + \left(\frac{2}{5}\right)\left(\frac{2}{3}\right)$$

$$= \frac{3}{15} + \frac{4}{15}$$

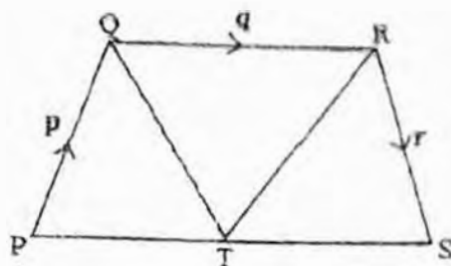
$$= \frac{7}{15}$$

Answer

(a) **Vimbai is likely to pass**

(b) (i) $\frac{2}{15}$

(ii) $\frac{7}{15}$



In the diagram $\overrightarrow{PQ} = p$, $\overrightarrow{QR} = q$ and $\overrightarrow{RS} = r$. Triangles PQT, QTR and TRS are equilateral.

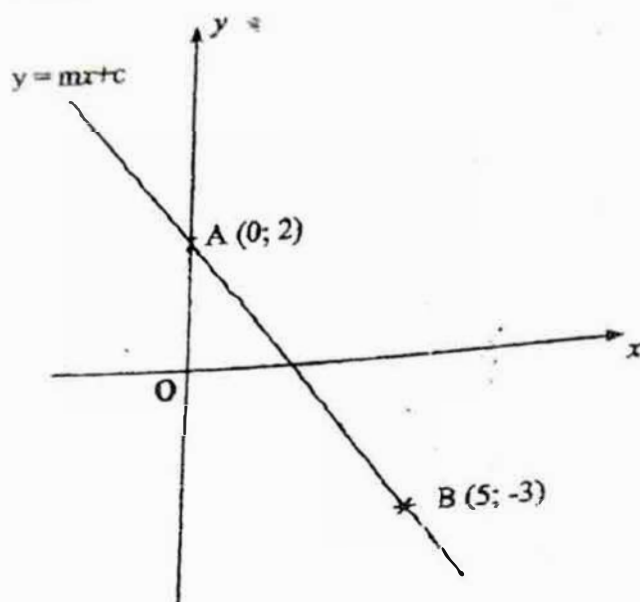
Express

- (i) $\overrightarrow{PS}$ in terms of
1. p , q and r ,
 2. q only.
- (ii) p in terms of q and r .

Suggested solution

$$\begin{aligned}
 23 \quad (i) \quad 1 \quad \overrightarrow{PS} &= \overrightarrow{PQ} + \overrightarrow{QR} + \overrightarrow{RS} \\
 2 \quad \overrightarrow{PT} &= \overrightarrow{TS} = \overrightarrow{QR} \quad (\text{by definition of equilateral } \Delta) \\
 \overrightarrow{PS} &= \overrightarrow{PT} + \overrightarrow{TS} \\
 &= \overrightarrow{PQ} + \overrightarrow{QR} \\
 &= 2q \\
 (ii) \quad \overrightarrow{PQ} &= \overrightarrow{PS} + \overrightarrow{SR} + \overrightarrow{RQ} \\
 p &= 2q - r - q \\
 p &= q - r
 \end{aligned}$$

$$\begin{aligned}
 \text{Answer} \quad (i) \quad 1. \quad \overrightarrow{PS} &= \overrightarrow{p} + \overrightarrow{q} + \overrightarrow{r} \quad (1) \\
 2. \quad 2\overrightarrow{q} & \quad (2) \\
 (ii) \quad \overrightarrow{q} - \overrightarrow{r} & \quad (2)
 \end{aligned}$$



The diagram shows the graph of $y = mx + c$, which passes through the points A (0; 2) and B (5; -3).

(a) Find the value of

(i) c ,

(ii) m .

(b) Calculate the length of AB, leaving your answer in surd form.

Suggested solution

24 (a) (i) $c = 2$

since c is the y intercept

(ii) $m = \frac{2 - (-3)}{0 - 5} = \frac{5}{-5} = -1$

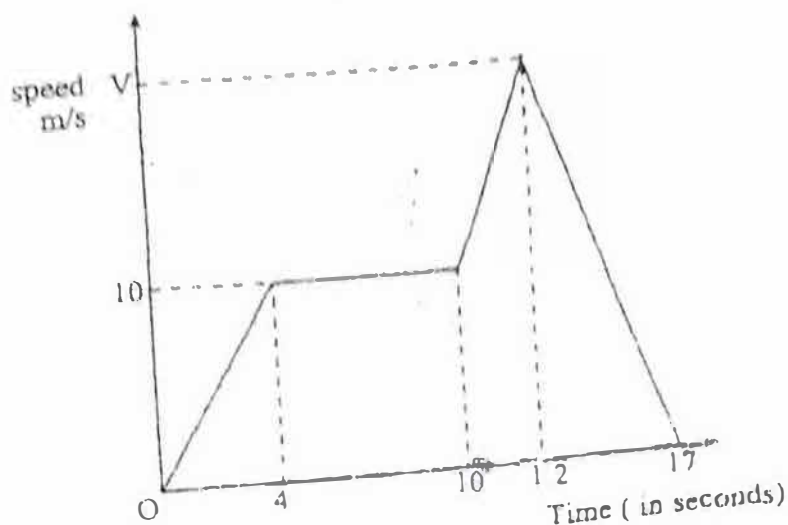
where m is the gradient

(b) Length AB = $|AB| = \sqrt{(0 - 5)^2 + (2 - (-3))^2}$
 $= \sqrt{(-5) \times (-5) + (2 + 3)^2}$
 $= \sqrt{25 + 25}$
 $= \sqrt{50}$
 $= \sqrt{25 \times 2}$
 $= \sqrt{25} \times \sqrt{2}$
 $= 5\sqrt{2}$

recall that $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$

Answer

- (a) (i) 2
 (ii) -1



The diagram represents the speed-time graph of a sprinter during an athletics training session.

- Calculate the distance the sprinter covers during the first 10 seconds.
- Given that the acceleration during the time interval from $t = 10$ to $t = 12$ is 5 m/s^2 , find the value of V .
- Calculate the deceleration of the sprinter, from $t = 12$ to the time the sprinter stops running.

Suggested solution

$$\begin{aligned}
 \text{(a)} \quad \text{Distance} &= \text{area of the trapezium} \\
 &= \frac{1}{2} \times (a + b) \times h \\
 &= \frac{1}{2} \times (6 + 10) \times 10, \\
 &= \frac{1}{2} \times 16 \times 10 \\
 &= 80 \text{ m}
 \end{aligned}$$

(ADDITION FIRST)

(b)

$$\text{Acceleration} = \frac{\text{Change in Velocity}}{\text{Change in Time}}$$

$$\text{Acceleration} = \frac{v - v_1}{t_2 - t_1} \quad \text{but Acceleration} = 5 \text{ m/s}^2 \text{ and } v = \text{velocity}$$

$$5 = \frac{v - 10}{12 - 10}$$

$$5 = \frac{v - 10}{2}$$

$$10 = v - 10$$

$$v = 20$$

i.e. velocity = 20 m/s

(c) Deceleration = $\frac{20 - 0}{17 - 12} = \frac{20}{5} = 4 \text{ ms}^{-2}$

Answer

(a)

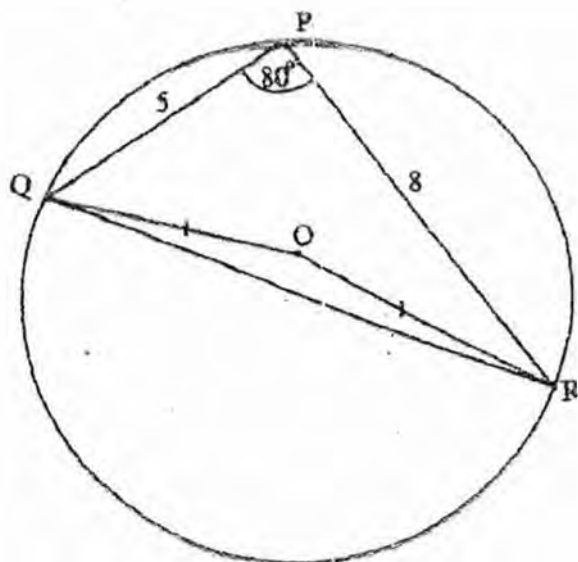
80 m

(b)

20 m/s

(c)

-4ms⁻²



The points P, Q and R lie on the circumference of a circle, centre O.
 $PQ = 5$ cm, $PR = 8$ cm and $\angle QPR = 80^\circ$.

Using as much of the information given as is necessary, calculate

- the area of triangle PQR,
- the value of QR^2 ,
- find the reflex $\angle QOR$.

$$[\sin 80^\circ = 0.985; \cos 80^\circ = 0.174; \tan 80^\circ = 5.67]$$

Suggested solution

26

(a)

$$\begin{aligned} \text{Area} &= \frac{1}{2} \times 5 \times 8 \sin 80^\circ \\ &= 20 \times 0.985 \\ &= 19.700 \end{aligned}$$

(b)

$$\begin{aligned} QR^2 &= 5^2 + 8^2 - 2 \times 8 \times 5 \times \cos 80^\circ, \\ &= 25 + 64 - 80(0.174) \\ &= 75.08 \end{aligned}$$

using the Cosine Rule

(c)

$$\text{reflex } \angle QOR = 360^\circ - 2(80^\circ) = 200^\circ$$

Using the theorem : angle subtended on the circumference is twice that subtended at centre

Answer

- | | |
|-----|--------|
| (a) | 19,700 |
| (b) | 75,08 |



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

MATHEMATICS

4028/2

PAPER 2

NOVEMBER 2013 SESSION

2 hours 30 minutes

Additional materials:

- Answer paper
- Geometrical instruments
- Graph paper (3 sheets)
- Mathematical tables
- Plain paper (1 sheet)

TIME 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **all** questions in **Section A** and any **three** questions from **Section B**.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

All working must be clearly shown. It should be done on the same sheet as the rest of the answer.

Omission of essential working will result in loss of marks.

If the degree of accuracy is not specified in the question and if the answer is not exact, the answer should be given to three significant figures. Answers in degrees should be given to one decimal place.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question. Mathematical tables or electronic calculators may be used to evaluate explicit numerical expressions.

This question paper consists of 13 printed pages and 3 blank pages.

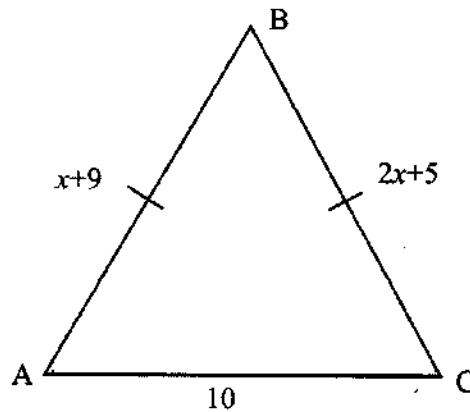
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Answer all the questions in this section.

- 4028/2 N2013

- 3 (a) Given that $m = 2p + 1$ and $n = p - 2$, express mn in terms of p in its simplest form. [2]

(b)



In the diagram, ABC is an isosceles triangle with $AB = BC$.

$AB = (x + 9)$ cm, $BC = (2x + 5)$ cm and the base, $AC = 10$ cm.

- (i) Form an equation in terms of x and solve it.
- (ii) Write down the length of BC .
- (iii) Calculate the area of triangle ABC .
- (iv) Given that all the lengths of the sides of $\triangle ABC$ were given to the nearest centimetre, calculate the **least** possible perimeter of the triangle. [9]

- 4 (a) If $dx = r + qx$,
- (i) find the value of d when $q = 3$, $r = -1$ and $x = 2$,
- (ii) express x in terms of d , q and r . [4]

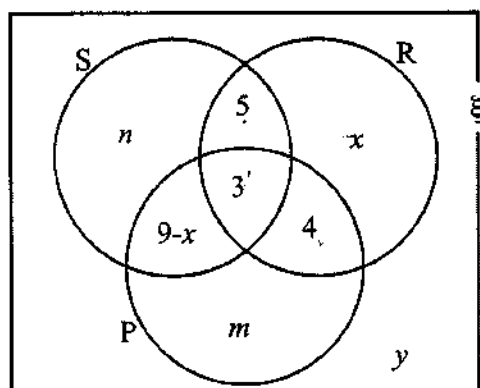
- (b) Fifty five people attending a workshop were asked to indicate which food item they liked, choosing from 'sadza', rice or potatoes.

S is the set of people who liked 'sadza', where $n(S) = 23$,

R is the set of people who liked rice, where $n(R) = 19$ and

P is the set of people who liked potatoes, where $n(P) = 12$.

The information was displayed in a Venn diagram and the numbers in each region represent the numbers of people who liked at least one of those food items.



- (i) Find the value of x .
- (ii) Calculate the number of people who liked 'sadza' only.
- (iii) Find the number of people who did **not** like any of the food items on offer. [6]

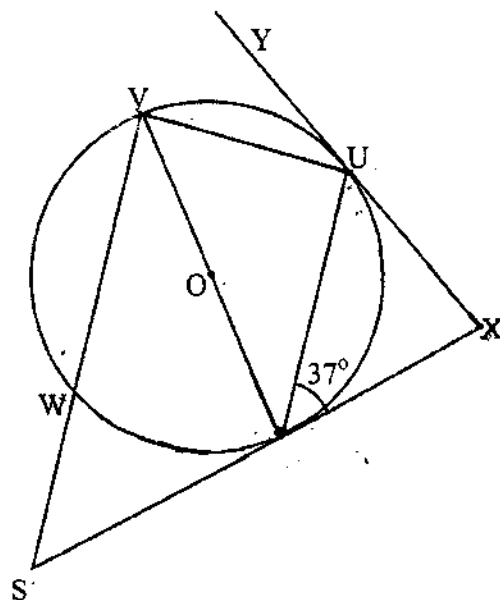
- 5 (a) Given that $P = \begin{pmatrix} 3 & -4 \\ 5 & 1 \end{pmatrix}$ and $Q = \begin{pmatrix} 2 \\ 8 \end{pmatrix}$,
 find

(i) the value of n if $PQ = \begin{pmatrix} 4n \\ 18 \end{pmatrix}$,

(ii) P^{-1} .

[4]

(b)



In the diagram, TUVW is a circle centre O. TOV is a diameter. STX and XUY are tangents to the circle at T and U respectively. SWV is parallel to TU and $\hat{UTX} = 37^\circ$

- (i) Find
1. $\hat{TSV}$,
 2. $\hat{S\hat{V}T}$,
 3. $\hat{U\hat{V}T}$,
 4. $\hat{TXU}$.

(ii) Name the triangle that is similar to $\triangle UVT$.

(iii) Given that $TU = 4,7$ cm,
 calculate the radius of the circle.

[8]

6 Answer the whole of this question on a sheet of plain paper provided.

Use ruler and compasses only and show clearly all construction lines and arcs.

(a) Construct on a single diagram,

- (i)** quadrilateral ABCD in which $AB = 6 \text{ cm}$, $\hat{ABC} = 120^\circ$, $BC = 7 \text{ cm}$,
 $CD = 9 \text{ cm}$ and $AD = 7 \text{ cm}$,
- (ii)** the locus of points equidistant from AB and AD,
- (iii)** the locus of points, inside the quadrilateral ABCD, that are 3 cm
from BC,
- (iv)** the shortest line from point C to AB produced.

[10]

**(b) Mark and label clearly the point P, inside the quadrilateral, which is
equidistant from AB and AD and 3 cm from BC.**

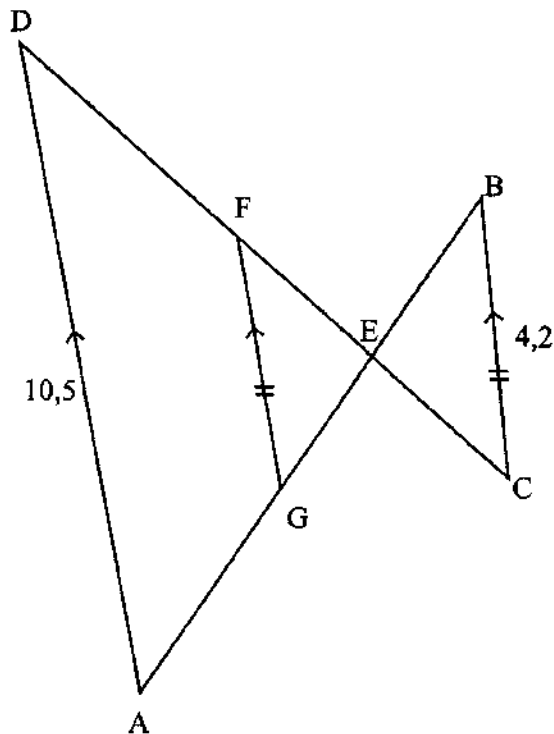
[1]

Answer any *three* questions in this section.

Each question carries 12 marks.

- 7 (a) Express $2 - 2\log 50$ as a logarithm of a single number. [2]

(b)

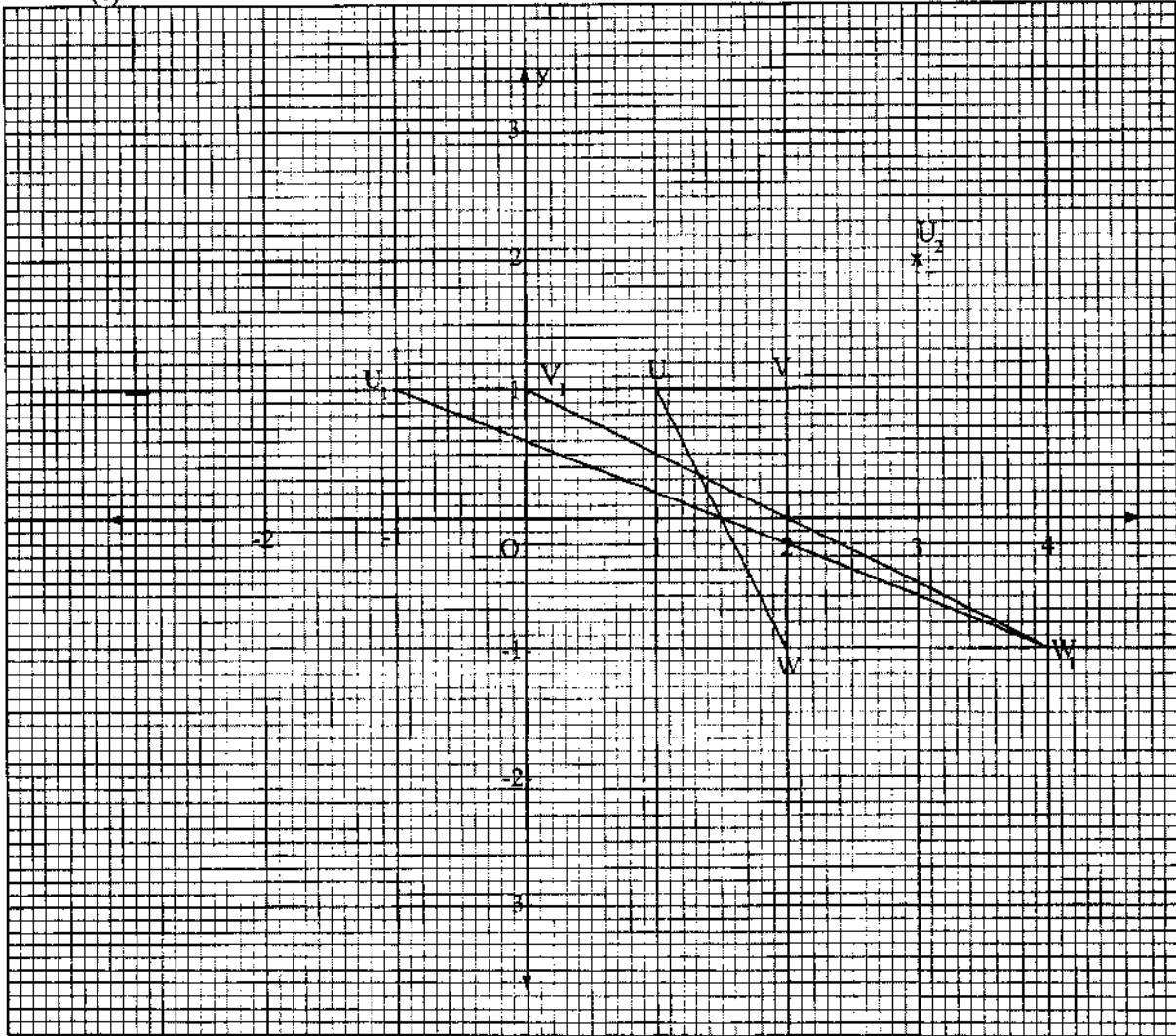


In the diagram, AB and CD intersect at E. the lines AD, GF and CB are parallel to each other. It is given that $BC = GF = 4,2$ cm and $AD = 10,5$ cm.

- (i) Describe fully the **single** transformation that maps $\triangle BCE$ onto $\triangle GFE$.
- (ii) $\triangle ADE$ is the image of $\triangle BCE$ under an enlargement.

State the centre and the scale factor of the enlargement. [5]

(c)



The diagram shows $\triangle UVW$, $\triangle U_1V_1W_1$ and point U_2 .

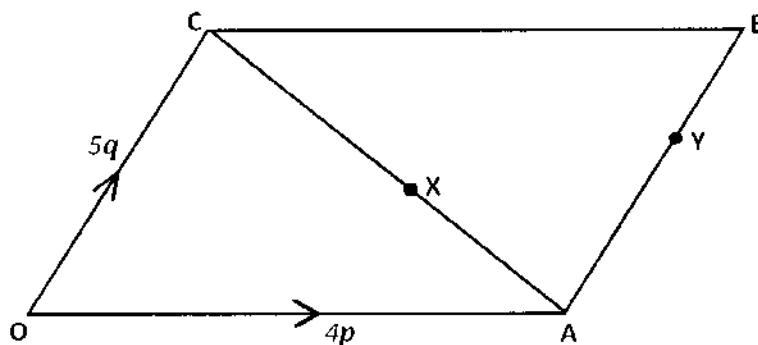
- (i) Describe completely the **single** transformation that maps $\triangle UVW$ onto $\triangle U_1V_1W_1$.
- (ii) Point U_2 is the image of point U under a two-way stretch.

- Find
1. the stretch factor with the y -axis invariant,
 2. the co-ordinates of V_2 , the image of V under this two-way stretch.

[5]

- 8 (a) If $\mathbf{g} = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$ and $\mathbf{h} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, express $\mathbf{g} + 2\mathbf{h}$ in the form $\begin{pmatrix} x \\ y \end{pmatrix}$. [2]

(b)



In the diagram, OABC is a parallelogram in which $\overrightarrow{OA} = 4\mathbf{p}$ and $\overrightarrow{OC} = 5\mathbf{q}$. X is a point on AC such that $AX:XC = 2:3$.

- (i) Express in terms of $\mathbf{p}$ and/or $\mathbf{q}$
 1. $\overrightarrow{AC}$,
 2. $\overrightarrow{OX}$ in its simplest terms.
- (ii) Y is a point on AB such that $\frac{AY}{AB} = k$, where k is a constant.
Express $\overrightarrow{OY}$ in terms of $\mathbf{p}$, $\mathbf{q}$ and k .
- (iii) Given that $OY = hOX$, where h is a constant,
write down another expression for $\overrightarrow{OY}$ in terms of $\mathbf{p}$, $\mathbf{q}$ and h .
- (iv) Using results in (ii) and (iii), find the value of h and the value of k .
- (v) Express $\frac{\text{the area of } \triangle OAY}{\text{the area of parallelogram OABC}}$, as a fraction in its simplest form.

[10]

- 9 Answer the whole of this question on a sheet of graph paper.

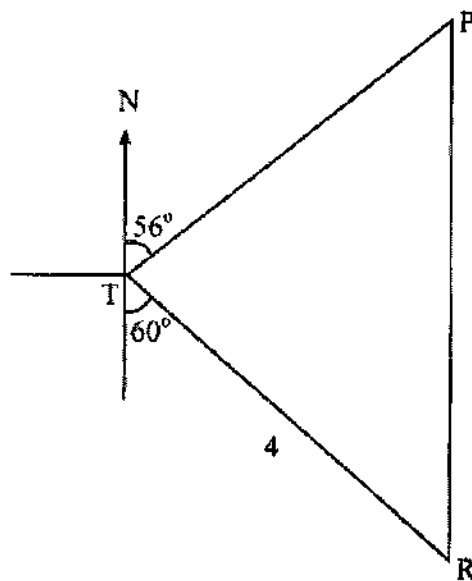
The following is an incomplete table of values for the function $y = 2 + \frac{5}{x}$.

x	1	2	3	4	5	6	7	8
y	7	p	3,7	3,3	3	q	2,7	2,6

- (a) Calculate the value of p and the value of q . [2]
- (b) Using a scale of 2 cm to represent 1 unit on each axis, draw the graph of $y = 2 + \frac{5}{x}$ for $1 \leq x \leq 8$. [4]
- (c) Use the graph to estimate
- the gradient of the graph of $y = 2 + \frac{5}{x}$ when $x = 2$,
 - the area enclosed by the graphs $y = 2 + \frac{5}{x}$, $x = 1$, $x = 2$ and the x -axis. [4]
- (d) On the same axes, draw the graph of $y = x$.
- Hence solve the equation $2 + \frac{5}{x} = x$. [2]

- 10 (a) Solve the equation $2x^2 - 4x - 3 = 0$, giving your answers correct to one decimal place. [5]

(b)



The diagram shows three points, P, R and T which are on level ground and $TR = 4$ km. From T, the bearing of P is $N 56^\circ E$, the bearing of R is $S 60^\circ E$ and P is due north of R.

(i) Calculate

1. the shortest distance between line PR and T,
2. PR.

(ii) From an aeroplane flying directly above point T the angle of depression of R is $20,3^\circ$.

Calculate the height of the aeroplane above the ground at T.

[7]

- 11 (a) The sum of interior angles of a polygon is $3\,240^\circ$. Three of its interior angles are 140° , 110° and 100° . The rest are all equal.

Find the size of each of the equal angles.

[4]

- (b) Forty pupils took part in a race and the distances to the nearest metre, that they covered in a certain time interval, are given in the frequency table below.

distance (in m)	$10 \leq x < 20$	$20 \leq x < 50$	$50 \leq x < 60$	$60 \leq x < 70$	$70 \leq x < 80$	$80 \leq x < 100$
frequency (f)	4	6	8	4	13	5
frequency density	0.4	p	0.8	q	r	0.25

- (i) State the modal class.
- (ii) If the information is to be represented on a histogram, find the values of p , q and r .
- (iii) Calculate the mean distance covered.
- (iv) Two of the pupils are selected at random to make a report on the race.

Find the probability that both pupils had covered 70 m or more in the race.

[8]

- 12 Answer the whole of this question on a sheet of graph paper.

Nomsa, a newspaper vendor, orders and sells two types of newspapers.
The Messenger and **The Arrival**.

- (a) Nomsa always orders at most 100 newspapers for resale daily.

If x represents the number of **The Messenger** and y the number of **The Arrival**, write down an inequality, in terms of x and/or y that satisfies the given condition.

[1]

- (b) She always sells more than 20 copies of **The Messenger** but the number of copies of **The Messenger** sold does not exceed double the number of copies of **The Arrival**.

Write down 2 inequalities involving x and/or y that satisfy these restrictions.

[2]

- (c) By the end of each day she ends up having sold more copies of **The Messenger** than of **The Arrival**.

Write down an inequality that satisfies this condition.

[1]

- (d) Using a scale of 2 cm to represent 20 newspapers on each axis, draw x -axis and y -axis each from 0 to 100.

Illustrate, by shading the **unwanted** regions, the region in which $(x : y)$ must lie.

[5]

- (e) It is given that **The Messenger** sells for \$1.20 and **The Arrival** sells for \$0,80 per copy.

Nomsa is given $2\frac{1}{2}\%$ commission on her sales.

Calculate the **highest** possible commission she could get.

[3]